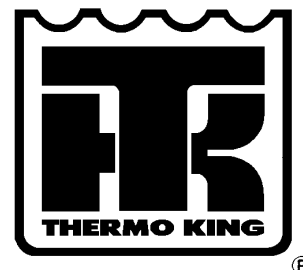




SERVICE BULLETIN



SUBJECT: Scroll Compressor Replacements

BULLETIN NO: C 059

DATE: January 21, 2003

Recently, there have been several scroll compressors replaced under warranty and returned to Thermo King for evaluation.

In testing done by TK and the vendor, these scroll compressors had no problems, indicating the attending technician did not use proper troubleshooting procedures before replacing the compressor.

Below are some points to check **BEFORE** replacing a scroll compressor thought to be defective.

1. Is main power line voltage correct?
2. Does the unit mode call for cooling? (Compressor LED ON)
 - a. Compressor will not run in heat, defrost modes or if no cooling required (Compressor LED OFF)
3. Are there any active alarms present?
 - a. Alarm 56 – Compressor Over Temp, possible causes:
 - Air in system
 - Low refrigerant charge
 - Failed compressor discharge sensor
 - Defective liquid injection valve
4. Are there any Error Messages present? (Controller will clear message if condition disappears)
 - a. Error messages are found in the Misc. Functions menu under STATUS, possible messages: (compressor may not be running but will come back on line when condition is corrected; some conditions delay start up to 10 minutes)
 - Message #7 – High pressure cutout, condenser sensor
 - Message #9 – High pressure cutout, condenser fan
 - Message #11 – Scroll compressor, high temperature
 - Message #12 – Scroll compressor, low pressure
5. Are high and low pressure switches are closed and OK?
6. Is the compressor contactor on the high voltage tray pulled in?
 - b. If it is and the compressor is not running, check the power cable connection at the contactor and the compressor.
 - c. Is compressor contactor wired correctly to ensure proper rotation of compressor
 - d. If not, is 29 VAC control circuit to contactor coil open.
WARNING: Ensure main power is disconnected prior to verification of high voltage connections.
7. Is compressor noisy? Possible causes:
 - a. Loose mounting bolts – tighten mounting bolts
 - b. Compressor rotating backwards – correct compressor contactor wiring

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